

Types of Research

I. Application Perspective

Theoretical (Basic) Research

- **Goal:** To expand knowledge without immediate practical application.
- **Example:** Investigating quantum mechanics principles.
- **Method:** Conceptual analysis, mathematical modeling, literature review.

Applied Research

- **Goal:** To solve specific, practical problems.
- **Example:** Developing a new app to improve mental health.
- **Method:** Combines theory with real-world testing.

II. Aims of the research

Three types of research—**exploratory**, **descriptive**, and **explanatory**

Exploratory Research

- **Purpose:** To investigate a problem that is not clearly defined.
- **Goal:** Discover ideas, ask questions, and identify patterns.
- **Approach:** Flexible and unstructured.
- **Example:** Interviewing people to understand why a new product isn't selling well.
- **Use Case:** Early stages of research when little is known about the topic.

Descriptive Research

- **Purpose:** To describe characteristics of a population or phenomenon.
- **Goal:** Answer questions like *who*, *what*, *when*, *where*, and *how*.
- **Approach:** Structured and often quantitative.
- **Example:** A survey measuring customer satisfaction across different age groups.
- **Use Case:** When you want to paint a detailed picture of a situation or group.

Explanatory Research (also called Causal Research)

- **Purpose:** To explain why something happens.
- **Goal:** Identify cause-and-effect relationships.
- **Approach:** Often experimental or longitudinal.
- **Example:** Studying whether social media usage causes anxiety in teenagers.
- **Use Case:** When you want to understand the underlying reasons or mechanisms.

Feature	Exploratory	Descriptive	Explanatory
Main Question	What's going on?	What is it like?	Why does it happen?
Structure	Unstructured	Structured	Highly structured
Data Type	Qualitative	Quantitative or mixed	Quantitative or mixed
Outcome	Hypotheses, insights	Profiles, patterns	Causal relationships

Each type builds on the other: **exploratory** research helps define the problem, **descriptive** research outlines the details, and **explanatory** research digs into the reasons.

Research approach to study a phenomena and determine Causal Relationships among different factors

It is done through **Experimental Research and Empirical Research**

Experimental Research

Experimental research is about **manipulating variables** to test a hypothesis.

- **Definition:** Involves controlled testing to establish **cause-and-effect** relationships.
- **Methodology:** Uses lab experiments, control groups, and randomization.
- **Environment:** Conducted in **controlled settings**, like laboratory.
- **Data Type:** Primarily **quantitative**.
- **Goal:** To **prove or disprove** a specific hypothesis.
- **Example:** Testing a new drug by giving it to one group and a placebo to another.

Feature	Empirical Research	Experimental Research
Approach	Observational	Manipulative
Control of Variables	Low	High
Setting	Natural or real-world	Controlled (e.g., lab)
Objective	Describe or infer	Establish causality
Data Types	Qualitative & Quantitative	Mostly Quantitative

So, in short: **empirical research observes in live situation**,
while **experimental research manipulates variables factors to see their effect**.

Empirical Research

- Empirical research is all about **observing the world as it is**. Observation and experiment are done in live environment. It seeks to answer questions using real-world evidence—whether through surveys, field studies, or experiments.
- **Definition:** Gathers knowledge through direct or indirect observation or experiment.
- **Methodology:** Includes surveys, case studies, field observations, and archival analysis.
- **Environment:** Typically conducted in **natural settings**.
- **Data Type:** Can be **qualitative** (e.g., interviews) or **quantitative** (e.g., statistics).
- **Goal:** To **describe, understand, or infer** patterns and relationships.
- **Example:** Studying consumer behavior by analyzing purchase data from retail stores.

Analytical Research:

At its core, analytical research is about **breaking down complex information**, identifying patterns, and using logic and evidence to reach conclusions. It's less about the method (observation vs. experimentation) and more about the **critical thinking and structured analysis** behind it.

The research can be Longitudinal or/ and cross-sectional.

Longitudinal Research

- **Goal:** To study changes over time.
- **Example:** Tracking academic performance of students over 10 years.
- **Method:** Repeated observations of the same subjects.

Causal-Comparative Research (Cross-section research)

- **Goal:** To determine cause-effect relationships by comparing groups.
- **Example:** Comparing test scores of students taught with different methods.
- **Method:** No manipulation—uses naturally occurring groups.

Research can be **Qualitative** Research and/or **Quantitative** Research

Qualitative Research	Quantitative Research
Data: text, images, audio	Data: numerical, labels
Data collection: unstructured, flexible Example: interview or survey with spontaneous open-ended questions	Data collection: structured, standardized, sometimes subjective Example: interview or survey with pre-defined close-ended questions
Usage: used in unexplored research topics and topics that are hard to quantify	Usage: used for precision and conclusive
Qualitative research uses a more unstructured and flexible approach, which works well in unexplored research fields	Quantitative research uses a more structured and standardized approach, which works well for zooming into details with high objectivity
In well-researched fields, quantitative research tends to be more common	

Example

Imagine a study testing whether listening to upbeat music improves creativity at work. Researchers split participants into two groups—one hears music, the other doesn't—and then measure their creative output. That's both **experimental** and **empirical**, because it involves:

- A controlled setup (experiment)
- Real-world data collection (empirical)

In short: **All experimental research is empirical**, but **not all empirical research is experimental**. Observational studies, for instance, are empirical but not experimental.

Experimental vs. Empirical vs. Analytical

Type of Research	Definition	Is Historical Data Used?	Causal Inference Possible?
Experimental	Involves controlled experiments where variables are manipulated to observe outcomes	Not typically (unless historical experiments were conducted)	Strong causal inference
Empirical	Based on observed and measured phenomena; uses data to test hypotheses	Yes, including historical data	With proper methods (e.g., regression, matching)
Analytical	Focuses on breaking down concepts, theories, or models through logical reasoning	Rarely uses raw data	Not designed for causal testing